8 Sampling Techniques and Statistical Inference



Pandurang Vasudeo Sukhatme
(July 27, 1911 - January 28, 1997)

Introduction

Pandurang Vasudeo Sukhatme (1911–1997) was an award-winning Indian statistician. He is known for his pioneering work of applying random sampling methods in agricultural statistics and in biometry, in the 1940s. He was also influential in the establishment of the Indian Agricultural Statistics Research Institute. As a part of his work at the Food and Agriculture Organization in Rome, he developed statistical models for assessing the dimensions of hunger and future food supplies for the world. He also developed methods for measuring the size and nature of the protein gap.

In any statistical investigation, the interest lies in the assessment of one or more characteristics relating to the individuals belonging to a group. When all the individuals present in the study are investigated, it is called complete enumeration, but in practice, it is very difficult to investigate all the individuals present in the study. So the technique of sampling is done which states that a part of the individuals are selected for the study and the assessment is made from the selected group of individuals. For example

(i) A housewife tastes a spoonful whatever she cooks to check whether it tastes good or not.
(ii) A few drops of our blood are tested to check about the presence or absence of a disease.
(iii) A grain merchant takes out a handful of grains to get an idea about the quality of the whole consignment.

These are typical examples where decision making is done on the basis of sample information. So sampling is the process of choosing a representative sample from a given population.

Learning Objectives

After studying this chapter students are able to understand

- sampling techniques
- random sampling
- simple random sampling
- stratified random sampling
- systematic sampling
- sampling and non-sampling errors
- sampling distribution
- statistical inference
- estimation
- test of statistical hypothesis

XEPSAT

8.1 Sampling

Sampling is the procedure or process of selecting a sample from a population. Sampling is quite often used in our day-to-day practical life.

XII Std - Business Maths & Stat EM Chapter 8.indd 175 2/29/2020 11:40:02 AM